

Here are your notes for "Logs (logarithms), Clearly Explained!!!".

Lecture Notes: Logs (Logarithms)

1. The Core Concept: Logs Isolate Exponents

The entire purpose of a logarithm is to **isolate the exponent** [01:55].

- **Rule:** The $\log$ base b of a number x asks the question: "What exponent do I need to raise b to in order to get x ?"
- **Example (Log Base 2):**
 - We know that $8 = 2^3$.
 - Therefore, $\log_2(8) = \log_2(2^3) = 3$.
 - The log function simply finds and returns the exponent (3).
- **Other Examples:**
 - $\log_2(4) = \log_2(2^2) = 2$ [02:15]
 - $\log_2(1) = \log_2(2^0) = 0$ [03:01]
 - $\log_2(0.5) = \log_2(1/2) = \log_2(2^{-1}) = -1$ [03:36]
 - $\log_2(0.25) = \log_2(1/4) = \log_2(2^{-2}) = -2$ [04:09]

Any number (even non-tidy ones like 5, 6, 7, or π) can be written as a power of 2, and the log will find that exponent [00:50].

2. Why Use a Log Scale? Symmetry & Fold Change

A log scale transforms a linear number line into a scale of exponents.

Linear Value	Written as Power of 2	$\log_2$ (Log Scale)
8	2^3	3
4	2^2	2
2	2^1	1
1	2^0	0
1/2 (0.5)	2^{-1}	-1
1/4 (0.25)	2^{-2}	-2
1/8 (0.125)	2^{-3}	-3

The Problem with Linear Scales:

On a linear scale, the distance from 1 to 8 (an 8-fold *increase*) is large. The distance from 1 to 1/8 (an 8-fold *decrease*) is very small [04:57]. This is not symmetrical.

The Benefit of Log Scales:

On a log scale, the distance from 0 (log of 1) to 3 (log of 8) is the same as the distance from 0 to -3 (log of 1/8). This makes it **the perfect way to visualize fold changes**, as it puts positive and negative fold changes on a symmetric scale [05:23], [05:59].

3. Special Cases and Applications

$\log(0)$ is Undefined

- There is no number x such that $2^x = 0$ [06:18].
- 2^{-1000} is a *tiny* positive number, but it's not zero [06:27].
- Programming languages like **R** often define $\log(0)$ as **-Infinity** (negative infinity), as the exponent would have to be infinitely negative to approach zero [06:53].

qPCR & Doubling Data

- Log base 2 is perfect for data that **doubles** at each step, like qPCR [07:26].
- An increase of "1 cycle" in the machine (doubling) is a change of +1 on the $\log_2$ scale [08:18].
- A difference of 3 cycles (8-fold increase) is a change of +3 on the $\log_2$ scale [08:34].

The Geometric Mean

- Taking the average of data on a log scale and then converting it back is called the **geometric mean** [09:27].
- **Example:** Data = [1, 2, 8]
 - **Arithmetic Mean:** $(1 + 2 + 8)/3 = 3.7$
 - **Geometric Mean:**
 - a. **Log Data** ($\log_2$): [0, 1, 3]
 - b. **Mean of Logs:** $(0 + 1 + 3)/3 = 1.3$
 - c. **Convert Back:** $2^{1.3} \approx 2.5$ [09:02]
- The geometric mean (2.5) is much less swayed by the outlier (8) than the arithmetic mean (3.7). It's more **robust to outliers** in data that grows exponentially [09:35].

4. Log Arithmetic: Simplifying Math

Because logs are exponents, they follow exponent rules.

Multiplication becomes Addition

- **Rule:** $\log(a \times b) = \log(a) + \log(b)$
- **Example:**
 - $\log_2(2 \times 4) = \log_2(8) = 3$
 - $\log_2(2) + \log_2(4) = 1 + 2 = 3$ [11:20]
- This is because $2^1 \times 2^2 = 2^{(1+2)}$. The log just isolates the added exponents.

Division becomes Subtraction

- **Rule:** $\log(a \div b) = \log(a) - \log(b)$
- **Example:**
 - $\log_2(2 \div 4) = \log_2(0.5) = -1$
 - $\log_2(2) - \log_2(4) = 1 - 2 = -1$ [12:45]

- This is because $2^1 \div 2^2 = 2^{(1-2)}$. The log isolates the subtracted exponents.

5. All Log Bases Work the Same

The principles of isolating exponents, symmetry, and arithmetic rules apply to **all log bases** [14:09].

- **Log Base 10** ($\log_{10}$): Useful for data that changes by powers of 10 (e.g., decibels, earthquake magnitudes) [14:09].
- **Natural Log** ($\ln$ or $\log_e$): Uses the number e (approx 2.7) as its base. This is often the default in programming (like R) but works the exact same way [14:28].
- **Log Base "Whatever"**: If your data triples at each step, you can use $\log_3$ [15:14].

http://googleusercontent.com/youtube_content/8